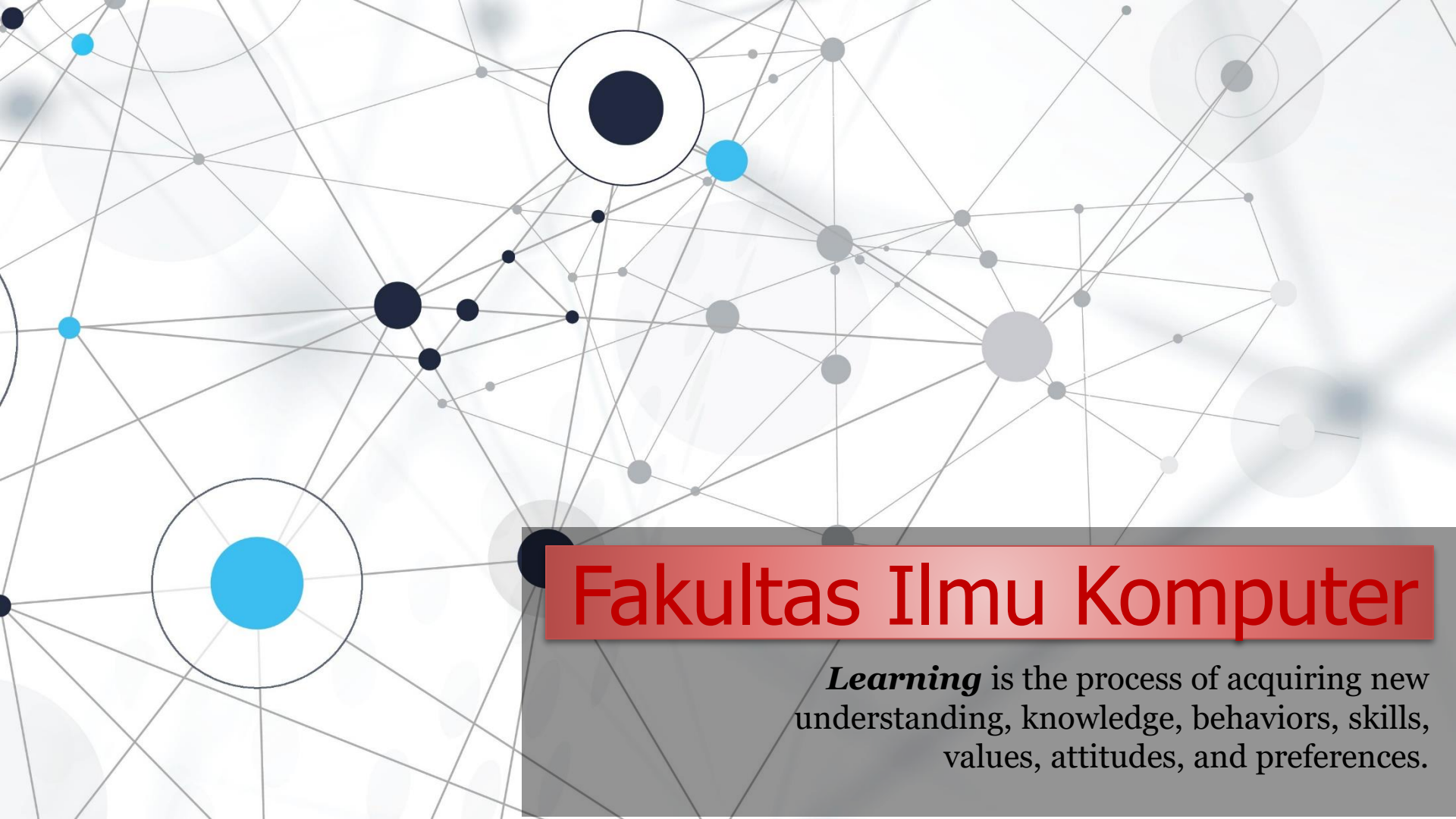


An abstract network diagram with various sized nodes (black, blue, grey) connected by thin grey lines. Some nodes are highlighted with larger circles. The background is white with faint grey circles.

Fakultas Ilmu Komputer

Learning is the process of acquiring new understanding, knowledge, behaviors, skills, values, attitudes, and preferences.

SSL Certificates for Security HTTPS Connections
(Secure Sockets Layer) = (HTTP to HTTPS)



Welcome to *the* **Back-web** **Development** *course*

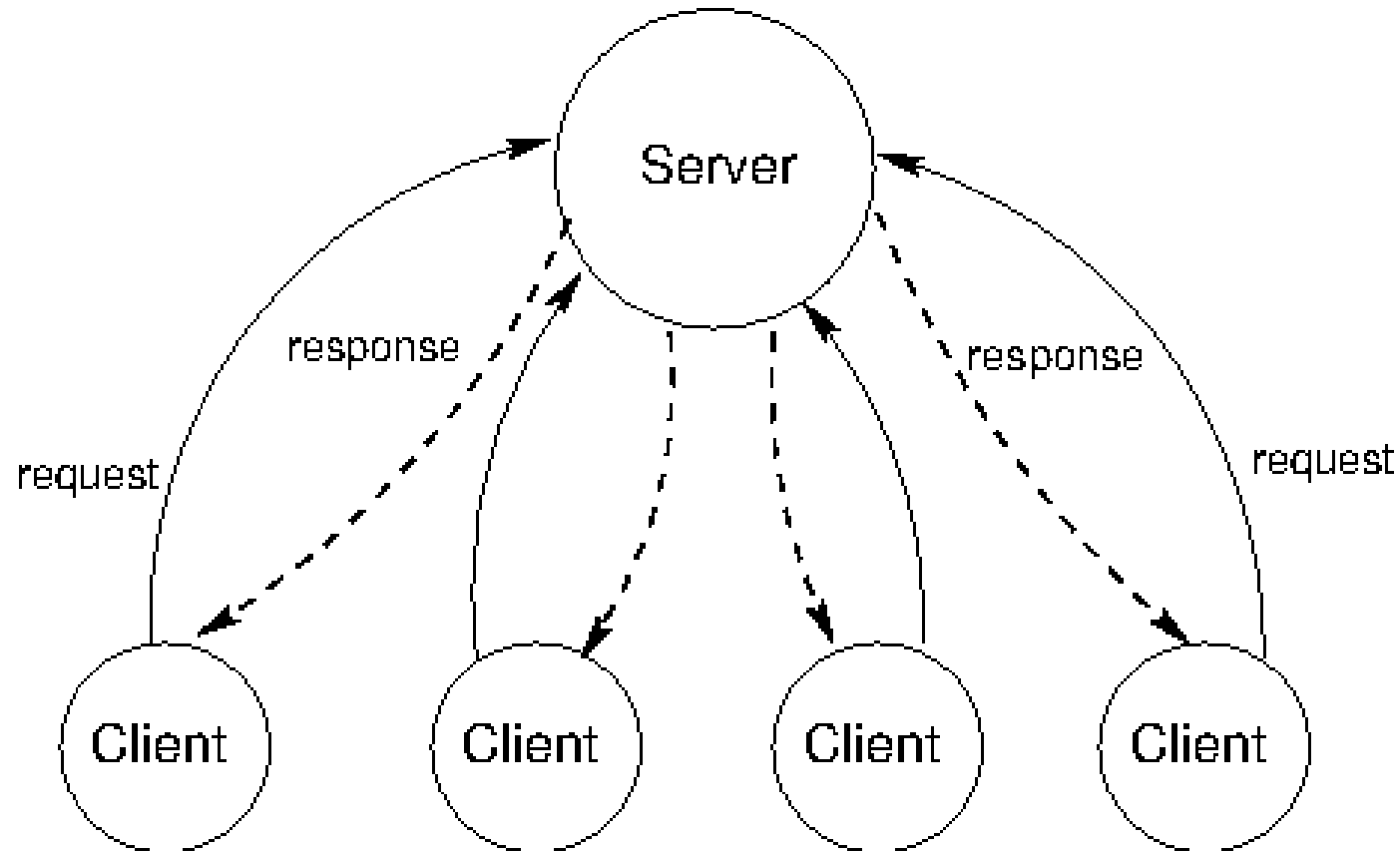
A decorative border of green leaves and branches at the top of the slide.

1 - TLS/SSL Implementation (HTTP *to* HTTPS)

A decorative border of green leaves and branches at the bottom of the slide.

Web Client & Web Server Communication

A **web client** is an application that communicates with a web server, using Hypertext Transfer Protocol (HTTP). The basic objective of the web server is to store, process and deliver web pages to the users. **Web browser is an example of a web client.** Basically, a **web server is a software that runs continuously and responds to requests** by other pieces of software, usually called a client.

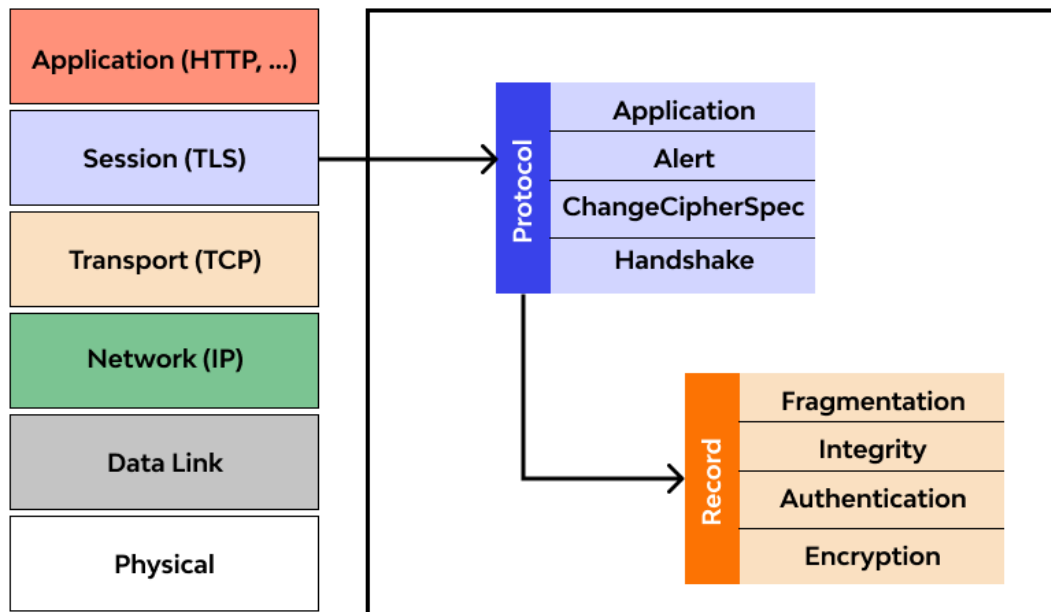




Security in the context of HTTP and HTTPS protocols, will inevitably involve the use of TLS (Transport Layer Security) and SSL (Secure Sockets Layer). The **implementation of TLS/SSL (HTTP to HTTPS) is essential to improve the security of web communication (client and server)**, especially when sensitive or private data must be transferred between the browser and the server, such as *login information, credit card details, customer data, government data* and others.

TLS (Transport Layer Security)

- ❖ Transport Layer Security (TLS) is a **cryptographic protocol designed to provide communications security** over a computer network. The protocol is widely used in applications such as email, instant messaging, and voice over IP, but its use in securing HTTPS remains the most publicly visible.
- ❖ TLS in web communication is a security protocol that **provides encryption and authenticity of data transferred over a network**. TLS replaces SSL and is designed to provide an additional layer of security on top of transport protocols such as TCP (Transmission Control Protocol).
- ❖ **TLS in web communication ensures that the data transferred between the user device and the server remains private and cannot be modified by unauthorized parties**. It involves the process of exchanging encryption keys and establishing a security session.



SSL (Secure Sockets Layer)

- ❖ SSL (Secure Sockets Layer) is an encryption-based Internet security protocol. It was first developed by Netscape in 1995 for the purpose of ensuring privacy, authentication, and data integrity in Internet communications.
- ❖ SSL is the predecessor (**pendahulu TLS**) to the modern TLS encryption used today. SSL in web communications was **the predecessor of TLS** and was used to provide similar security in transferring data over a network.
- ❖ SSL in web communications uses symmetric and asymmetric cryptography to protect data and provide trust in the identity of the server and, in some cases, in the client.



Secure Sockets Layer

HTTP to HTTPS

HTTP (Hypertext Transfer Protocol) is a protocol used to transfer data on the web. However, HTTP does not provide adequate security (*tidak menyediakan keamanan yang memadai*) because the **data transmitted between the web browser and the server is not encrypted**, making it vulnerable to attacks and monitoring by third parties.

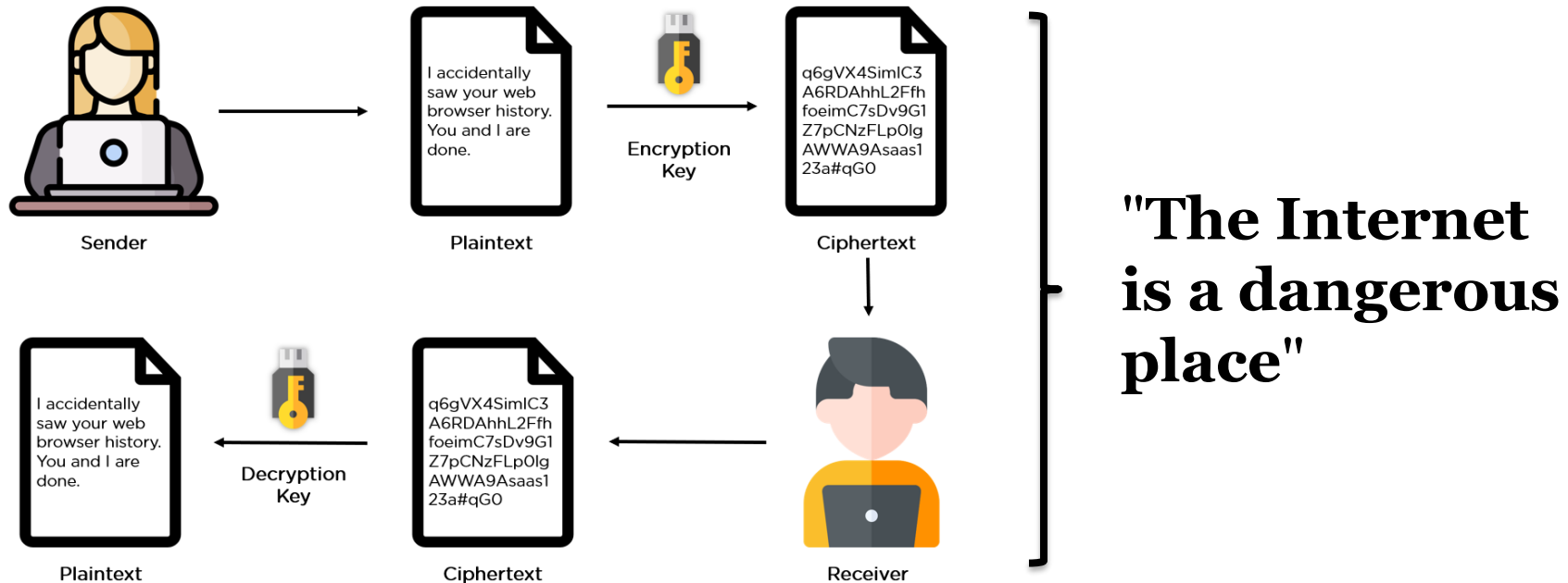


HTTPS (Hypertext Transfer Protocol Secure) is a secure version of HTTP. In HTTPS, the **data transferred between the web browser and the server is encrypted** using a security protocol, such as TLS or SSL, to protect the confidentiality and **integrity of the data**.



Mengapa komunikasi ini perlu dijaga?

Many websites today, especially those that provide **financial services**, **e-commerce**, or **collect personal information**, invest substantial resources into maintaining the security. These systems will undoubtedly use security protocols such as HTTPS to encrypt communication between users and servers, implement firewalls and intrusion detection systems, and regularly conduct security audits.



Internet dapat menjadi lingkungan yang berisiko karena adanya potensi ancaman keamanan seperti malware, phishing, and other cyber crimes. Selain itu, internet juga dapat menjadi tempat di mana informasi pribadi dapat terancam keamanannya (compromised).

A decorative border of green leaves and branches at the top of the slide.

2 – Why “HTTP *to* HTTPS” important *for* Back-end developer?

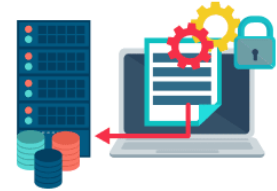
A decorative border of green leaves and branches at the bottom of the slide.



Data science analysis



Business logic development



Secure data storage

BACKEND DEVELOPER RESPONSIBILITIES



Web and mobile app backend development



System architecture building



Payment process configuration

It is important for web developers on both the **front-end and back-end sides** to understand the concepts of HTTP and HTTPS as well as SSL certificates because these are related to security and data integrity on the web. By understanding the concepts of HTTP, HTTPS, and SSL certificates, web developers (*website and server side*) can optimize security and user confidence in the website being built. This is also in line with industry trends and evolving security requirements.

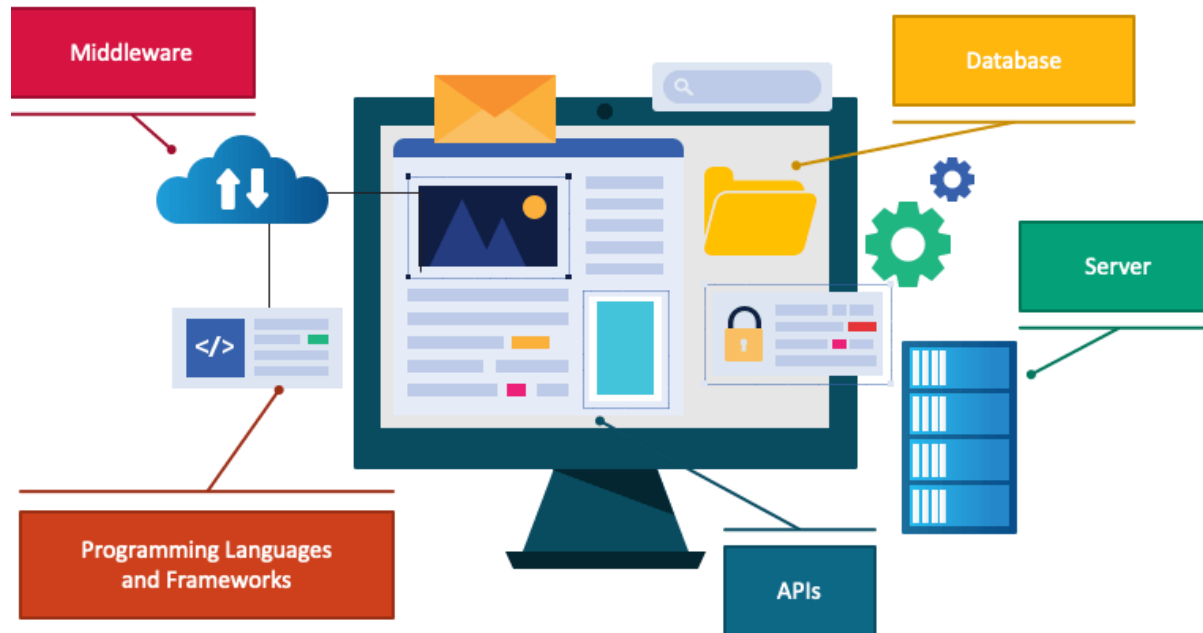
Collaboration between Dev & Ops

DevOps encourages collaboration between development and operations teams. In the context of transitioning to HTTPS, **developers are responsible for implementing secure coding practices**, while operations teams handle the configuration and maintenance of the web servers. Collaboration ensures a holistic approach to security and performance.



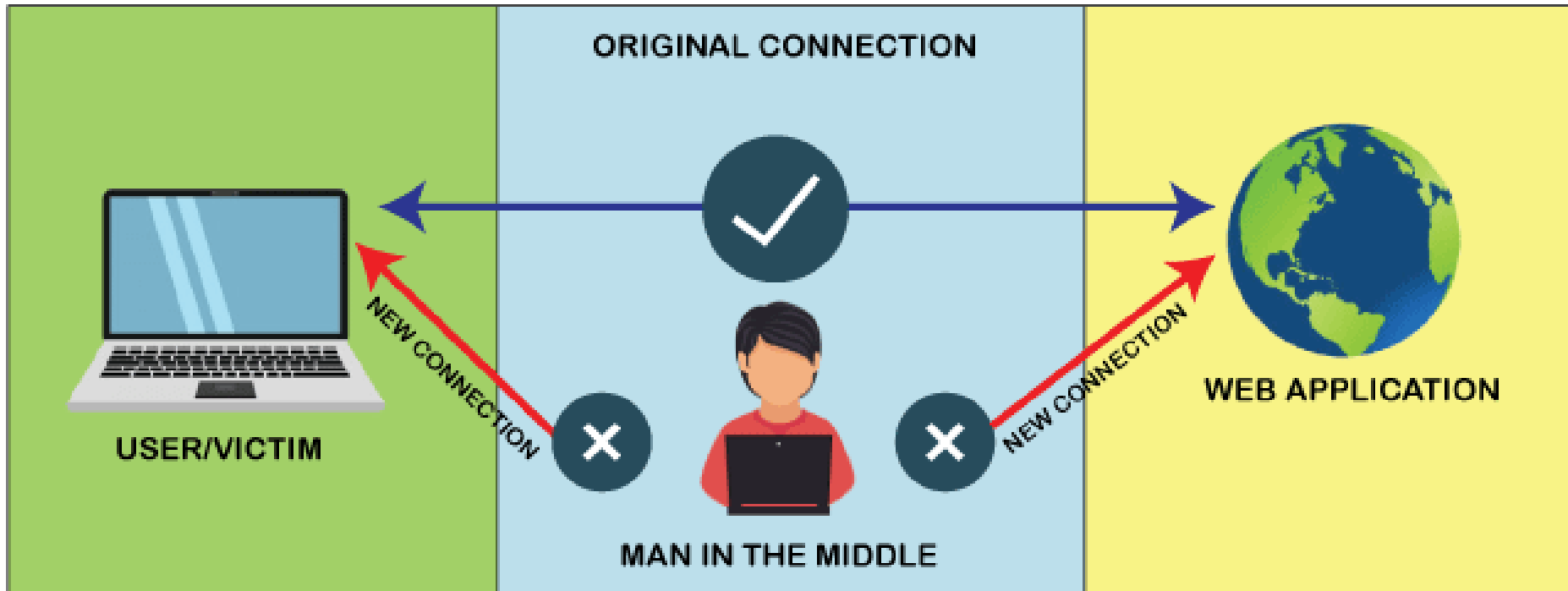
DevOps practices often **emphasize security as an integral part of the development lifecycle**. Transitioning from HTTP to HTTPS is a security measure to encrypt data in transit, providing a secure communication channel between the web client and server. DevOps teams may be involved in implementing and managing the HTTPS configuration to ensure compliance with security standards.

User Data Security



- ❖ HTTPS (Hypertext Transfer Protocol Secure) uses SSL/TLS (Secure Sockets Layer/Transport Layer Security) to **encrypt the data sent between the server and the user**. Thus, sensitive information such as passwords, credit card information, and personal data of the user remains safe from attacks by hackers trying to steal or manipulate the data.
- ❖ Modern users tend to be more trusting of websites that use HTTPS. Modern browsers often mark HTTP sites as insecure, while HTTPS sites are given a mark indicating that the connection is secure. This can increase user trust in your website.
- ❖ Search engines, such as Google, give higher rankings to websites that use HTTPS. Therefore, using an SSL certificate and building your website with the HTTPS protocol can provide advantages in terms of SEO.
- ❖ SSL certificates help **prevent man-in-the-middle attacks**, where an **attacker tries to spy** on or alter the data sent between the user and the server.

Data Integrity (No Data manipulation)



- ❖ **Man in the Middle (MiTM)** is a *cyberattack* carried out to steal information and spy on the victim. MiTM perpetrators (Pelaku MiTM) will **take advantage of insecure internet connections** to carry out their actions. MiTM will position itself between the victim and the website being used. Without realizing the victim is being observed by hackers.
- ❖ SSL certificates (HTTP to HTTPS connections) not only encrypt data, but also provide data integrity. This means that the data sent and received cannot be manipulated by third parties during the transmission process.



website



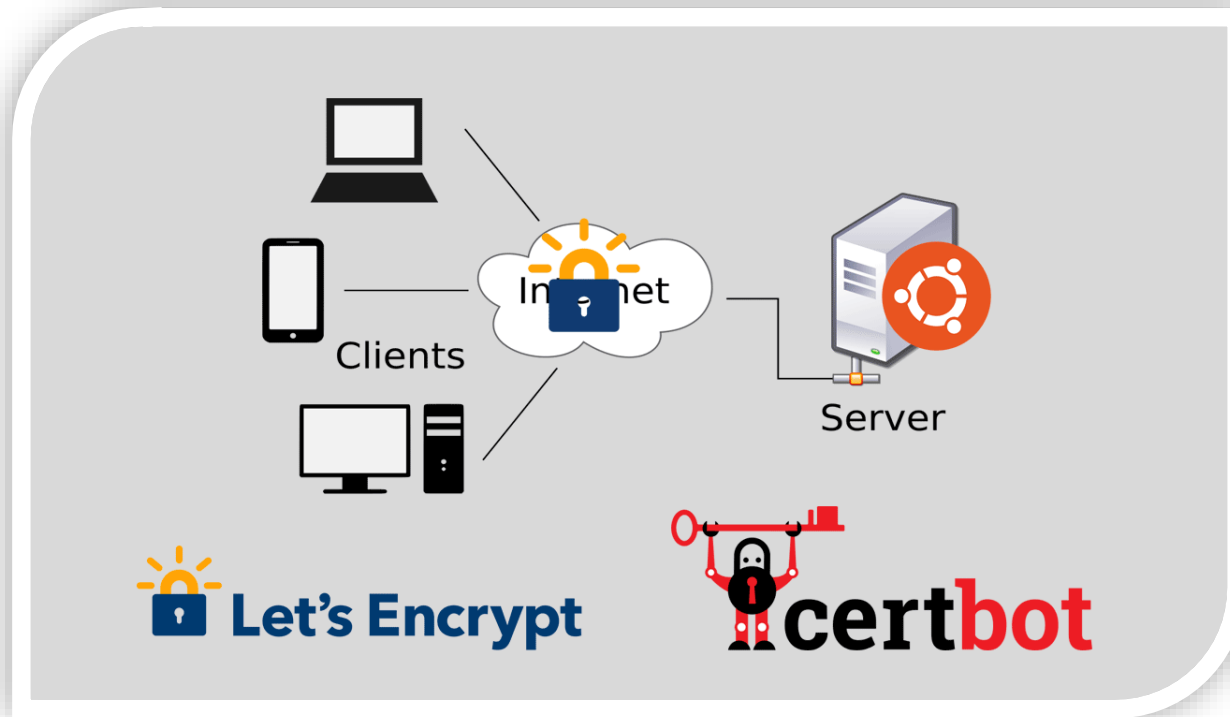
https://www.

Connection is secure

■ Certificate (Valid)

SSL certificates (Secure Sockets Layer) are what enable websites to use HTTPS, which is more secure than HTTP. An SSL certificate is a data file hosted in a website's origin server. SSL certificates make SSL/TLS encryption possible, and they contain the website's public key and the website's identity, along with related information.

Let's Encrypt SSL Certificate



Let's Encrypt is a free, automated, and open certificate authority. Let's Encrypt is a global Certificate Authority (CA) to let people and organizations around the world obtain, renew, and manage SSL/TLS certificates. These certificates can be used by websites to enable secure HTTPS connections. Let's Encrypt offers Domain Validation (DV) certificates.

Common Requirement SSL Certificates

- 1) Requires a **DOMAIN NAME** to secure with an SSL certificate. Let's Encrypt provides certificates for public domain names that are accessible over the internet.
- 2) Requires access to the **SERVER** where the domain is hosted. This can be SSH access or any other type of access that allows us to manage the server configuration.
- 3) Need to have a **WEB SERVER** running on the server. Apache, Nginx, and several other web servers are supported by Let's Encrypt.
- 4) Need to have **CERTBOT (ACME CLIENT)** or another ACME (Automatic Certificate Management Environment) client installed on the server. Certbot is the official tool supported by Let's Encrypt to automate SSL certificate management. Ensure that the server's operating system supports Certbot and the latest version of Let's Encrypt. Ubuntu 22.04 should support Certbot well.



Exercise *for* Students

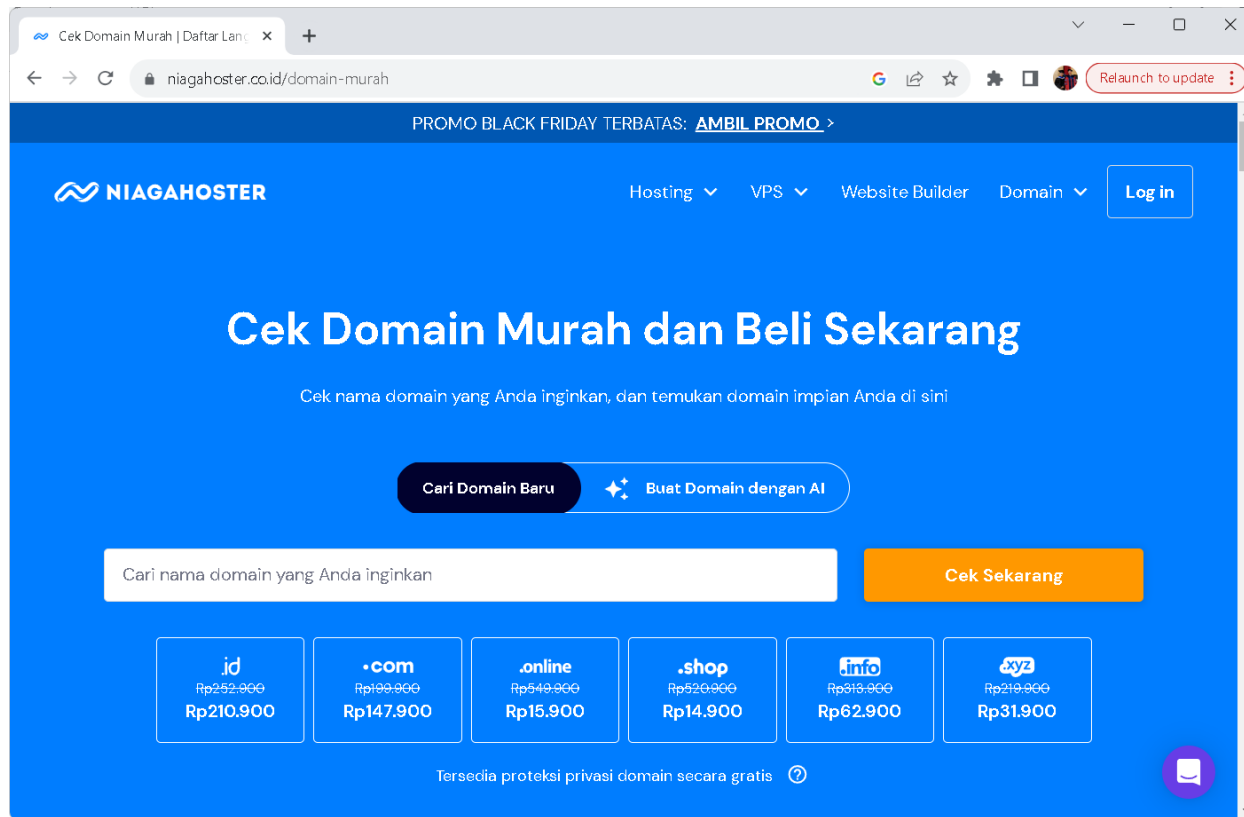


Exercise #1

(Buat domain name murah di niagahoster.co.id)

Open & Sign In NiagaHoster

<https://www.niagahoster.co.id/domain-murah>



The screenshot shows the NiagaHoster website interface. At the top, there's a navigation bar with the NiagaHoster logo, links for Hosting, VPS, Website Builder, and Domain, and a Log in button. A promotional banner for Black Friday is visible. The main heading is 'Cek Domain Murah dan Beli Sekarang'. Below this, there's a subtext: 'Cek nama domain yang Anda inginkan, dan temukan domain impian Anda di sini'. There are two buttons: 'Cari Domain Baru' and 'Buat Domain dengan AI'. A search input field is present with the placeholder text 'Cari nama domain yang Anda inginkan'. To the right of the input field is a 'Cek Sekarang' button. Below the input field, there are six domain options displayed in boxes:

Domain	Original Price	Current Price
.id	Rp252.000	Rp210.900
.com	Rp199.000	Rp147.900
.online	Rp549.000	Rp15.900
.shop	Rp520.000	Rp14.900
.info	Rp813.000	Rp62.900
.xyz	Rp219.000	Rp31.900

At the bottom, there's a note: 'Tersedia proteksi privasi domain secara gratis' with a question mark icon. A chat icon is visible in the bottom right corner.

Cari Domain Murah

Cek Domain Murah dan Beli Sekarang

Cek nama domain yang Anda inginkan, dan temukan domain impian Anda di sini!

[Cari Domain Baru](#) [Buat Domain dengan AI](#)

[Cek Sekarang](#)



Cari “semmywebsite.online”

Cek Domain Murah | Daftar Lanjut

niagahoster.co.id/domain-murah

NIAGAHOSTER

Hosting VPS Website Builder Domain [Log in](#)

[Hasil Pencarian](#) Semua Ekstensi

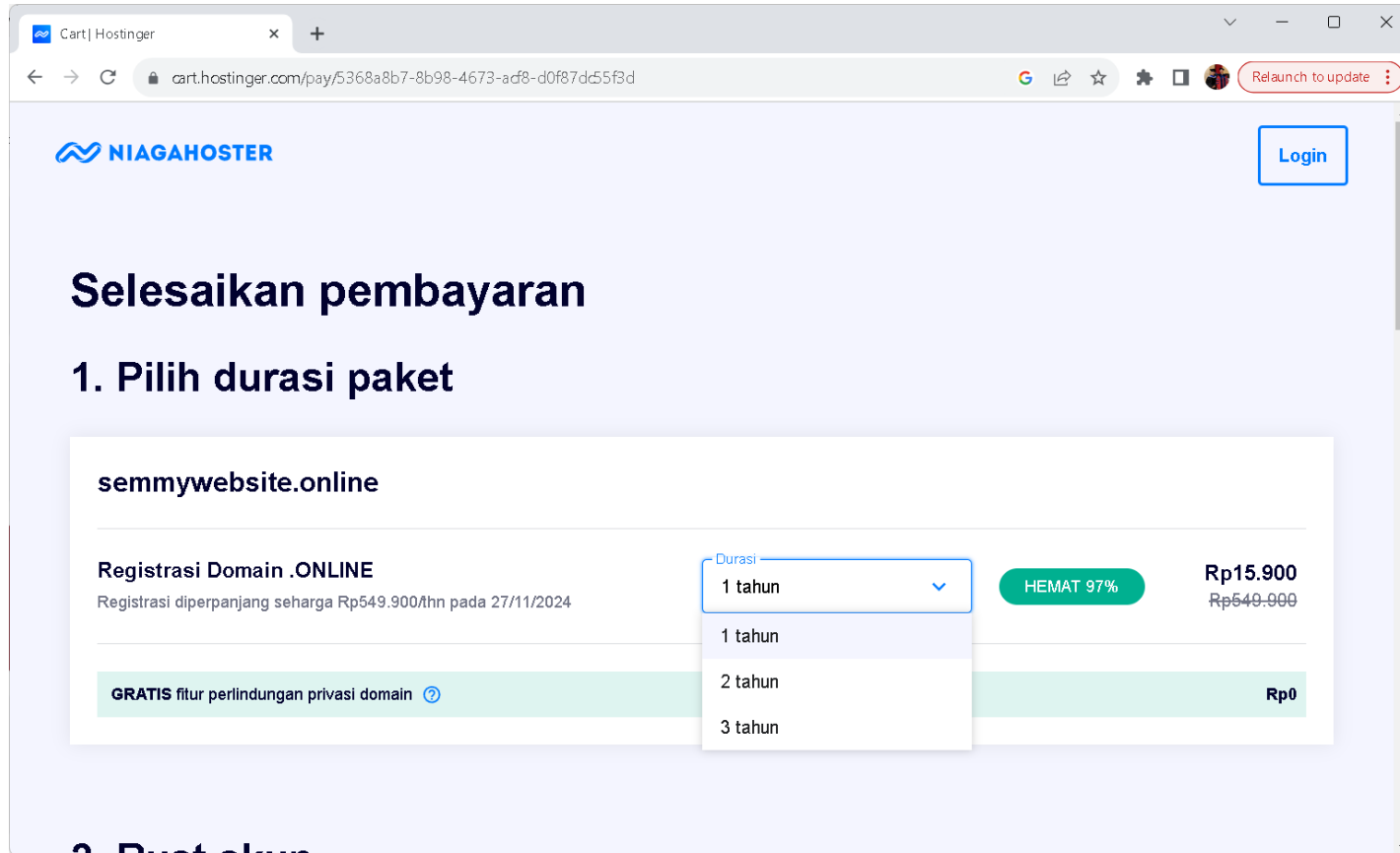
Selamat! Domain yang Anda Cari Tersedia

✓ **semmywebsite.online** **HEMAT 97%** ~~Rp549.900~~ **Rp15.900/thn** [Pilih](#)

This domain is suitable for creating a personal or professional website, as it conveys a sense of being a semi-official or semi-formal online presence.

Click button ini untuk memilih domain name.

Durasi Registrasi Domain



The screenshot shows the Niagahoster website interface. At the top, there's a navigation bar with the Niagahoster logo and a 'Login' button. The main heading is 'Selesaikan pembayaran' (Complete payment). Below it, the first step is '1. Pilih durasi paket' (1. Choose package duration). The domain 'semmywebsite.online' is entered. The registration details show 'Registrasi Domain .ONLINE' with a note about a 97% discount. A dropdown menu for 'Durasi' (Duration) is open, showing options for 1, 2, and 3 years. The current selection is 1 year, which is priced at Rp15.900, down from a regular price of Rp549.900. A green banner indicates 'GRATIS fitur perlindungan privasi domain' (Free domain privacy protection feature).

Cart | Hostinger

cart.hostinger.com/pay/5368a8b7-8b98-4673-adf8-d0f87d55f3d

Relaunch to update

NIAGAHOSTER

Login

Selesaikan pembayaran

1. Pilih durasi paket

semmywebsite.online

Registrasi Domain .ONLINE
Registrasi diperpanjang seharga Rp549.900/thn pada 27/11/2024

Durasi

- 1 tahun
- 1 tahun
- 2 tahun
- 3 tahun

HEMAT 97%

Rp15.900
~~Rp549.900~~

GRATIS fitur perlindungan privasi domain

Rp0

2. Buat akun

Metode Pembayaran

Cart | Hostinger

← → ↺

cart.hostinger.com/pay/5368a8b7-8b98-4673-adf8-d0f87dc55f3d

🔍 📄 ⚙️ 🛒 🧑 Relaunch to update

3. Pilih pembayaran

☒ Kartu Kredit

☐ BCA VA

☐ Mandiri Virtual Account

☐ BNI Virtual Account

☐ BRI Virtual Account

☐ CIMB Niaga Virtual Account

☐ Maybank Virtual Account

☐ Permata Bank Virtual Account

☐ QRIS

☐ GoPay

☐ OVO

☐ Indomaret

semmywebsite.online

✓ Registrasi domain - 1 tahun

Rp549.900 **Rp15.900**

✓ Proteksi Privasi Domain WHOIS

Rp0

Negara

Indonesia

Kabupaten/provinsi (opsional)

Kota (opsional)

Alamat (opsional)

Kode pos (opsional)

Diskon Paket - 97%

-Rp534.000

Pajak & Biaya Tambahan

Rp4.868

Total

Rp554.768 **Rp20.768**

Punya Kode Kupon?

Buat Tagihan Pembayaran

Pembayaran Aman dan Terenkripsi

Dengan checkout, Anda setuju dengan [Ketentuan Penggunaan](#) kami dan mengonfirmasi bahwa Anda telah membaca [Kebijakan Privasi](#) kami. Anda dapat membatalkan biaya perpanjangan layanan kapan saja.

Masuk Ke Dashboard Niagahoster

Click di tab Domains



The screenshot displays the Niagahoster dashboard interface. At the top, a navigation bar includes the Niagahoster logo and links for Home, Hosting, Emails, Domains (which is highlighted with a blue underline), VPS, and Billing. On the right side of the navigation bar, there are links for Pro Panel, a search icon, a US flag, a calendar icon, a help icon, and a user profile icon. A red notification bubble in the top right corner says 'Relaunch to update'. On the left sidebar, the 'Domains' tab is selected and highlighted in blue. Below it are links for 'Get a New Domain' and 'Transfers'. The main content area is titled 'Domains' and features two primary action cards: 'Get a new domain' with a magnifying glass icon and 'Transfer an existing domain' with a double arrow icon. Below these cards is a section titled 'My domains' which contains a promotional banner for 'unkpresent.com' with a 'SAVE 26%' badge and a 'Get now' button. At the bottom of the 'My domains' section is a search bar and a table header with columns for 'Domain Name', 'Status', 'Expires At', and 'Auto-renewal'. A 'Give feedback' link is located in the bottom left corner of the sidebar.

Cek Status Domain Name

The screenshot shows the hPanel Domains management interface. A red arrow points to the 'Domains' tab in the top navigation bar. Another red arrow points to the 'Domains' link in the left sidebar. A third red arrow points to the 'Manage' button for the domain 'unkpresent.site' in the table below.

Domains | hPanel

hpanel.hostinger.com/domains

Home Hosting Emails **Domains** VPS Billing

Pro Panel **BETA**

Domains

- Get a New Domain
- Transfers

Get a new domain
Find a perfect domain name using instant search with a wide range of available extensions

Transfer an existing domain
Bring domain to Hostinger from another registrar to ensure easy management from a single place

My domains

Protect your brand. Get **unkpresent.com** now! [More options...](#) **SAVE 26%** ~~Rp199.999~~ **Rp147.900/1st yr** [Get now](#)

Search

☐	Domain Name	Status	Expires At	Auto-renewal	
☐	unkpresent.site	Active	2024-08-08	☐	Manage

Click button "Manage"

Pastikan Sudah Verified by Email

The screenshot shows the 'Domain Overview' page in the hPanel interface. The domain 'unkpresent.site' is listed with a status of 'Active' and an 'Email verification status' of 'Verified'. The expiration date is '2024-08-08' with a 'Renew' button. The transfer authorization code is masked with asterisks and an eye icon to toggle visibility. The nameservers are 'ns1.dns-parking.com' and 'ns2.dns-parking.com' with a 'Change' button. Below the domain information, there are two sections: 'Auto-renewal' which is currently 'DISABLED' with an 'Enable auto-renewal' button, and 'Forward your domain to an existing website' which has a 'Forward domain' button. The interface includes a top navigation bar with links to Home, Hosting, Emails, Domains, VPS, and Billing, and a sidebar with a 'Give feedback' link.

Domain Overview | hPanel

hpanel.hostinger.com/domain/unkpresent.site/domain-overview

Relaunch to update

NIAGAHOSTER Home Hosting Emails **Domains** VPS Billing

Pro Panel BETA

Domain Information

Status	Active
Email verification status	Verified
Expires at	2024-08-08 Renew
Transfer authorization code	***** Toggle visibility
Nameservers	ns1.dns-parking.com ns2.dns-parking.com Change

Auto-renewal DISABLED

We will automatically renew the domain to keep it active

[Enable auto-renewal](#)

Forward your domain to an existing website

All of your traffic will be redirected to your entered website.

[Forward domain](#)

[Give feedback](#)

Click Tab “DNS/Nameservers”

The screenshot shows the Niagahoster hPanel interface for managing DNS records for the domain `unkpresent.site`. The left sidebar contains navigation options: **Domain Overview**, **DNS / Nameservers** (selected), and **Domain Ownership**. The main content area is titled **Manage DNS records** and includes a form to add new records and a table of existing records.

Manage DNS records

These records define how your domain behaves. Common uses include pointing your domain at web servers or configuring email delivery for your domain.

Add Record Form:

- Type: A
- Name: @
- Points to:
- TTL: 14400
- Add Record**

DNS Records Table:

Type	Name	Priority	Content	TTL		
A	www	0	27.112.78.66	14400	Delete	Edit
A	@	0	27.112.78.66	14400	Delete	Edit

A red arrow points to the **Content** column of the table, indicating where to change the IP address.

Change with your public IP Address



ASSIGNMENT

Exercise #2

(Install Let's Encrypt SSL Certificate on Ubuntu 22.04)

Install Certbot Client

- 1) Updates the package lists:
 - **sudo apt-get update**
- 2) Install the **Certbot client** with the Apache plugin:
 - **sudo snap install --classic certbot**
 - **sudo apt-get install certbot python3-certbot-apache**
 - **sudo apt-get install certbot --fix-missing**
 - **sudo apt-get install certbot python3-certbot-apache apache2**
- 3) Test if the Certbot client was installed correctly:
 - **sudo certbot --help**
- 4) Obtain certificate for domain name:
 - **sudo certbot --apache**
- 5) Add (Multiple) New domain name:
 - **sudo certbot --apache -d presensi.unklab.ac.id**
- 6) Delete old certificate (sertifikat ssl lama):
 - **sudo certbot delete**

Sesudah perintah ini dijalankan, anda akan diminta untuk memasukkan email dan domain name yang telah dibuat.

Renew Certificate (after 3 months)

1) Certbot is on Autorenewal:

- `sudo systemctl list-timers`
- `sudo systemctl enable certbot.timer`
- `sudo systemctl start certbot.timer`
- `sudo systemctl status certbot.timer`
- `sudo certbot renew --dry-run`

2) Cek Status Certificate:

- `sudo certbot certificates`

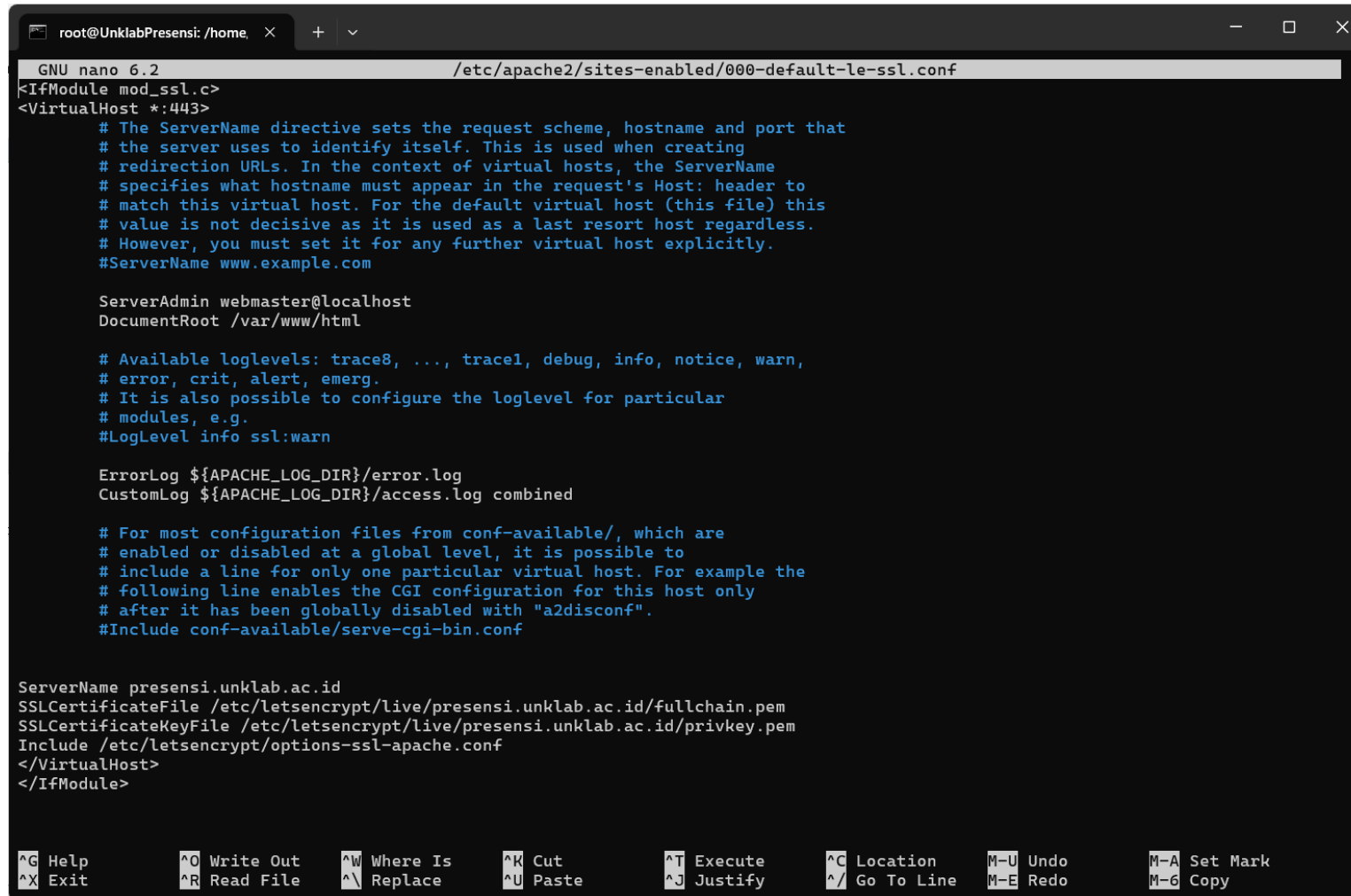
3) Renew Certificate:

- `sudo certbot renew`

Check Configuration 443

Certbot is on Autorenewal:

➤ `sudo nano /etc/apache2/sites-enabled/000-default-le-ssl.conf`



```
root@UnklabPresensi: /home. x + v
GNU nano 6.2 /etc/apache2/sites-enabled/000-default-le-ssl.conf
<IfModule mod_ssl.c>
<VirtualHost *:443>
    # The ServerName directive sets the request scheme, hostname and port that
    # the server uses to identify itself. This is used when creating
    # redirection URLs. In the context of virtual hosts, the ServerName
    # specifies what hostname must appear in the request's Host: header to
    # match this virtual host. For the default virtual host (this file) this
    # value is not decisive as it is used as a last resort host regardless.
    # However, you must set it for any further virtual host explicitly.
    #ServerName www.example.com

    ServerAdmin webmaster@localhost
    DocumentRoot /var/www/html

    # Available loglevels: trace8, ..., trace1, debug, info, notice, warn,
    # error, crit, alert, emerg.
    # It is also possible to configure the loglevel for particular
    # modules, e.g.
    #LogLevel info ssl:warn

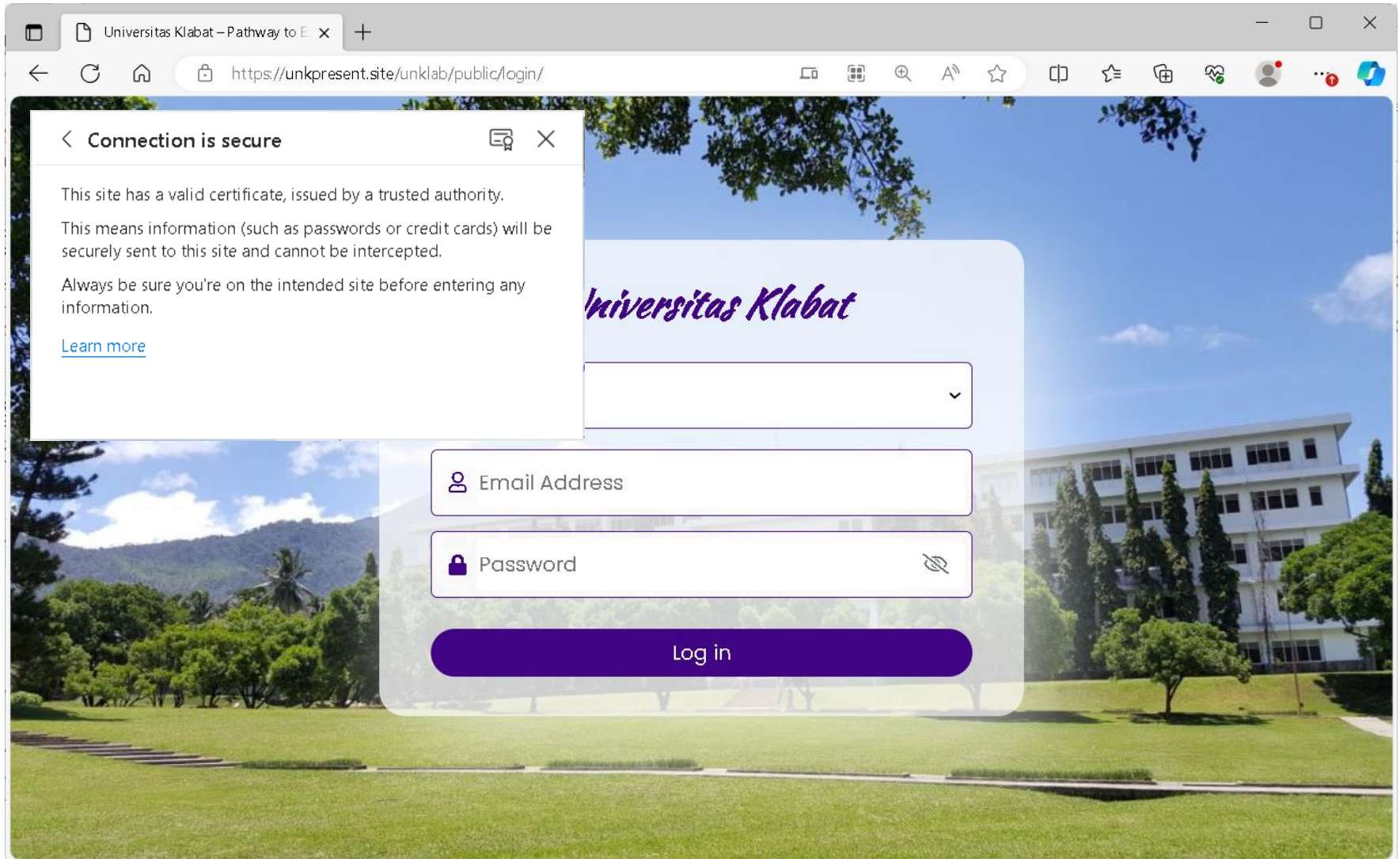
    ErrorLog ${APACHE_LOG_DIR}/error.log
    CustomLog ${APACHE_LOG_DIR}/access.log combined

    # For most configuration files from conf-available/, which are
    # enabled or disabled at a global level, it is possible to
    # include a line for only one particular virtual host. For example the
    # following line enables the CGI configuration for this host only
    # after it has been globally disabled with "a2disconf".
    #Include conf-available/serve-cgi-bin.conf

    ServerName presensi.unklab.ac.id
    SSLCertificateFile /etc/letsencrypt/live/presensi.unklab.ac.id/fullchain.pem
    SSLCertificateKeyFile /etc/letsencrypt/live/presensi.unklab.ac.id/privkey.pem
    Include /etc/letsencrypt/options-ssl-apache.conf
</VirtualHost>
</IfModule>

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo     M-A Set Mark
^X Exit      ^R Read File  ^_ Replace    ^U Paste      ^J Justify    ^/_ Go To Line  M-E Redo     M-6 Copy
```


Check Https Access





Exercise #3

(Redirect Public IP (80 HTTP) ke Domain (443 HTTPS))

Check File Apache Config Aktif

Tujuannya adalah supaya akses lewat IP (contoh: <http://103.xxx.xxx.xxx>) otomatis pindah ke domain dengan HTTPS, berikut adalah step-by-step di server Apache2:

#1 Cara cek file config aktif. Jalankan:

`sudo apache2ctl -S`

```
root@ISC2025:/home/iscadmin# sudo apache2ctl -S
AH00558: apache2: Could not reliably determine the server's fully qualified
domain name, using 10.50.189.59. Set the 'ServerName' directive globally to
suppress this message
VirtualHost configuration:
*:443             isc.unklab.ac.id (/etc/apache2/sites-enabled/000-defa
ult-le-ssl.conf:2)
*:80              10.50.189.59 (/etc/apache2/sites-enabled/000-default.
conf:1)
ServerRoot: "/etc/apache2"
Main DocumentRoot: "/var/www/html"
Main ErrorLog: "/var/log/apache2/error.log"
Mutex watchdog-callback: using_defaults
Mutex rewrite-map: using_defaults
Mutex ssl-stapling-refresh: using_defaults
Mutex ssl-stapling: using_defaults
Mutex ssl-cache: using_defaults
Mutex default: dir="/var/run/apache2/" mechanism=default
Mutex mpm-accept: using_defaults
PidFile: "/var/run/apache2/apache2.pid"
Define: DUMP_VHOSTS
```

Redirect Permanent

#2 Edit file 000-default.conf (default file ini saja sudah cukup):

```
sudo nano /etc/apache2/sites-available/000-default.conf
```

#3 Isi jadi seperti ini (Ganti public IP address di bawah ini):

```
<VirtualHost *:80>
```

```
ServerAdmin webmaster@localhost
```

```
ServerName 113.193.158.86
```

```
ServerAlias isc.unklab.ac.id www.isc.unklab.ac.id
```

```
Redirect permanent / https://isc.unklab.ac.id/
```

```
ErrorLog ${APACHE_LOG_DIR}/error.log
```

```
CustomLog ${APACHE_LOG_DIR}/access.log combined
```

```
</VirtualHost>
```

Dengan begini, semua request ke port 80 (baik pakai IP maupun domain) langsung di-redirect ke <https://isc.unklab.ac.id/>.

Reload Apache

#4 Simpan lalu reload Apache:

```
sudo systemctl reload apache2
```

(Optional) Tambahkan config khusus supaya semua request selain domain diarahkan. Kalau kita ingin semua host/ip apapun yang diarahkan ke server ini tetap dialihkan ke domain utama, bisa pakai:

```
<VirtualHost *:80>  
    ServerName default  
    Redirect permanent / https://isc.unklab.ac.id/  
</VirtualHost>
```

#5 Reload Apache

```
sudo systemctl reload apache2
```

Output

The screenshot shows a web browser with a search bar containing '103.176.79.218'. The search results show a link to '12ISC 2025 - 103.176.79.218' and a '404 - Not Found' error. A red overlay box contains the following instructions:

- Akses ke <http://103.192.178.91>**
→ langsung redirect ke <https://isc.unklab.ac.id/>
- Akses ke <http://isc.unklab.ac.id>**
→ langsung naik ke HTTPS <https://isc.unklab.ac.id/>
- Akses ke <http://www.isc.unklab.ac.id>**
→ ikut redirect ke HTTPS

The background of the screenshot shows a website banner for 'Advancing Science, Driving Digital Evolution' with buttons for 'SUBMISSION' and 'REGISTRATION'.

END PRESENTATION

Thank you for your attention

Instructor: S – W – T

THANK YOU

